TOPIC 01 · EXCEL FUNDAMENTALS WITH COPILOT
LAB 2 THEME · FORMULAS & XLOOKUP

Lab 2 — Generate and Verify Formula Columns and XLOOKUPs for Job Requisitions

Maps to	LO1: use Copilot to write XLOOKUP and calculated columns from plain-English rules, explain them, and reconcile them to independent control formulas
Duration	35 minutes
Tools	Excel · Copilot (Allow editing, Chat only) · XLOOKUP · structured references · BrightPath_Job_Requisitions.xlsx
Mock data	data/BrightPath_Job_Requisitions.xlsx
Course	Microsoft 365 Copilot for Excel (C34) · v3.2
Prompts	https://tertiarycourses.github.io/C34-Microsoft-365-Copilot-for-Excel/prompts/#lab-2

Goal

Add recruiter names, days open, fill rate, estimated placement fee and an ageing flag to a requisitions table, then prove each column is right.

What You Will Do

BrightPath's permanent-recruitment desk tracks 60 client job orders. You will ask Copilot to add an XLOOKUP for the recruiter's name and four calculated columns written from business rules, ask it to explain each formula, hand-check one requisition and reconcile column totals to independent formulas on the Control sheet. You will then ask Copilot which recruiter generated the most estimated fees and verify the answer.

What You Will Build

BrightPath_Job_Requisitions.xlsx with Recruiter_Name, Days_Open, Fill_Rate, Est_Fee and Ageing_Flag columns in tblRequisitions, overdue requisitions highlighted, and a Control sheet that reconciles the totals

Prerequisites

- Lab 1 completed, or Copilot access confirmed on a local .xlsx file.
- This lab folder's data/BrightPath_Job_Requisitions.xlsx.

Data note

Use only the supplied fictional BrightPath data or information you are authorised to process. Do not paste credentials, confidential business material or real employees' personal data into Copilot, rules or skills.

Steps

Step 1. Copy BrightPath_Job_Requisitions.xlsx from this lab's data folder into a work-c34 folder on your computer and open it in Excel for Microsoft 365 (desktop). Work on this local copy - no OneDrive upload is needed. Check that Formulas > Calculation Options is Automatic, then open Copilot from the Copilot button on the Home tab or the floating Copilot icon at the bottom right. Read the About_This_Data sheet before you prompt. Note the fee rule: Est_Fee = Filled x Monthly_Salary_Budget x 12 x Fee_Rate, and the report date in Control!B1.

PROMPT / CHECK — copy and adapt

```
Tables : tblRequisitions (60 rows) | tblRecruiters (6 rows)
Control!B1 Report_Date = 2026-06-30
```

Step 2. Choose Allow editing. Ask Copilot to add the lookup column, then ask it to explain the formula and show one row next to the matching row in tblRecruiters.

PROMPT / CHECK — copy and adapt

```
In tblRequisitions, add a column Recruiter_Name that looks up Recruiter_Name from
tblRecruiters using Recruiter_ID. Use XLOOKUP and return "Not found" if there is no match.

Then: Explain the Recruiter_Name formula, and show one row's Recruiter_ID and Recruiter_Name
next to the matching row in tblRecruiters.
```

Step 3. Ask for Days_Open with an explicit rule for open requisitions. Read the explanation, then ask Copilot to show an open row (blank Close_Date) and a closed row so you can confirm the rule.

PROMPT / CHECK — copy and adapt

```
Add a column Days_Open to tblRequisitions. If Close_Date is blank, use Control!B1 minus
Open_Date; otherwise use Close_Date minus Open_Date. Show whole numbers.

FOLLOW-UP: Show one open requisition (blank Close_Date) and one closed requisition with
Open_Date, Close_Date and Days_Open, and the subtraction behind each.
```

Step 4. Ask for Fill_Rate and Est_Fee, one prompt each. Format Fill_Rate as a percentage and Est_Fee as S$ with no decimals. Then ask Copilot to show the calculation for JOB-2002: Filled 4 x S$4,900 x 12 x 25% = S$58,800.

PROMPT / CHECK — copy and adapt

```
Add a column Fill_Rate = Filled divided by Openings, formatted as a percentage.

Add a column Est_Fee = Filled x Monthly_Salary_Budget x 12 x Fee_Rate, formatted as S$ with no
decimals.
Explain the formula.

FOLLOW-UP: Show the Est_Fee calculation for JOB-2002 step by step.
```

Step 5. Ask for the ageing flag and a highlight, then ask Copilot how many requisitions are flagged.

PROMPT / CHECK — copy and adapt

```
Add a column Ageing_Flag that shows "Overdue" when Status is "Open" and Days_Open is greater
than 45, otherwise leave it blank. Highlight rows where Ageing_Flag is "Overdue" with a light
amber fill.

FOLLOW-UP: How many rows have Ageing_Flag = "Overdue"?
```

Step 6. Ask Copilot to build independent controls on the Control sheet with the exact formulas below. The independent formulas recalculate from the raw columns, so they do not rely on Copilot's new columns. Each pair must match.

PROMPT / CHECK — copy and adapt

```
Enter these labels and formulas on the Control sheet exactly as written. Then tell me each
result and whether each pair (B3 and B4, B6 and B7) matches:
A3 Total Est_Fee (column) B3 =SUM(tblRequisitions[Est_Fee])
A4 Total fee (independent) B4 =SUMPRODUCT(tblRequisitions[Filled],tblRequisitions[Monthly_Sal
ary_Budget],tblRequisitions[Fee_Rate])*12
A5 Avg days to fill (Filled) B5
=AVERAGEIFS(tblRequisitions[Days_Open],tblRequisitions[Status],"Filled")
A6 Overdue (independent) B6
=COUNTIFS(tblRequisitions[Status],"Open",tblRequisitions[Days_Open],">45")
A7 Overdue flags (column) B7 =COUNTIF(tblRequisitions[Ageing_Flag],"Overdue")
```

Step 7. In Chat only, ask which recruiter generated the most estimated fees. Then ask Copilot (Allow editing) to add the SUMIFS check to the Control sheet and to log all prompts on Prompt_Log.

PROMPT / CHECK — copy and adapt

```
CHAT ONLY:
Using tblRequisitions, which Recruiter_Name has the highest total Est_Fee? Show the total for
every recruiter.

CHECK (Allow editing):
On the Control sheet, add this check for the top recruiter and tell me if it matches your
answer:
=SUMIFS(tblRequisitions[Est_Fee],tblRequisitions[Recruiter_Name],"<name>")

LOG (Allow editing):
Prompt_Log already lists every prompt in this lab. For each one I ran, fill in What Copilot
did and How I verified, and set Decision to Keep. I will tell you any row to change to Revise
or Undo.
```

Sample Prompts

Use these prompts in the Copilot pane (or adapt them). Name the table and columns, state the output, and always verify the result. The same prompts are in sample-prompts.txt in this lab folder.

Copy-ready prompts online:
https://tertiarycourses.github.io/C34-Microsoft-365-Copilot-for-Excel/prompts/#lab-2

#	Purpose	Prompt
1	Lookup	Add a column Recruiter_Name that looks up Recruiter_Name from tblRecruiters using Recruiter_ID with XLOOKUP. Return "Not found" if missing.
2	Duration	Add a column Days_Open: if Close_Date is blank use Control!B1 minus Open_Date, otherwise Close_Date minus Open_Date.
3	Ratio	Add a column Fill_Rate = Filled / Openings formatted as a percentage.
4	Business rule	Add a column Est_Fee = Filled x Monthly_Salary_Budget x 12 x Fee_Rate, formatted as S$ with no decimals.
5	Flag	Add a column Ageing_Flag that shows "Overdue" when Status is Open and Days_Open > 45, otherwise blank.
6	Explain	Explain the formula in the selected cell step by step, and tell me how blanks are handled.
7	Summary row	Add a total row to tblRequisitions that sums Openings, Filled and Est_Fee and averages Days_Open.
8	Question	Which recruiter has the highest total Est_Fee? Show every recruiter's total, highest first.
9	Several steps	Add Fill_Rate and Est_Fee columns, add a total row, then tell me which recruiter has the highest total Est_Fee.

Test It

Check	Expected result
Total estimated fees (Control B3 and B4)	Both S$1,030,512
Average days to fill (B5)	34.6 days (29 filled requisitions)
Overdue requisitions (B6 and B7)	Both 13
Sample row JOB-2002	Est_Fee S$58,800, Days_Open 36
Recruiter_Name lookup	Every row has a name - no "Not found"
Top recruiter by fees	Aisha Rahman - S$221,760

Checkpoint

Keep the workbook in work-c34. Lab 3 uses its own metrics workbook in the Lab 3 folder.

Troubleshooting

If this happens	Fix
Days_Open is negative or huge for open requisitions.	The formula used today's date or a blank as zero. Ask Copilot to use Control!B1 explicitly, or edit the formula.
Est_Fee is 12 times too small or too large.	Re-read the rule: fee is on first-year salary (monthly x 12). Ask Copilot to explain the formula and correct it.
The new column is outside the table.	Click inside tblRequisitions before prompting and name the table in the prompt; or drag the table's resize handle.
Copilot cannot add a formula column.	Confirm Calculation Options is Automatic and the data is an Excel table; if it still fails, save, close and reopen the workbook.

Challenge

Ask Copilot to add a Fee_Achieved column that is Est_Fee for Filled requisitions only, then build a small summary of Fee_Achieved by Sector with SUMIFS. Check that the sector totals add up to the grand total. Then bring in other data: select + (Add work content) and attach Lab 1's BrightPath_Candidate_Pipeline.xlsx (Copilot can only attach files it can find in your OneDrive or SharePoint, so skip this part if you work locally only), and ask: 'Add a sheet Sector_View that compares open requisitions by Sector in this workbook with hires by Client_Sector in the attached pipeline.' Check one sector's hires with COUNTIFS in the pipeline workbook.

Reflection

Which of today's formulas would have looked correct but been wrong without a hand-check, and how did you catch it?